

ML-Agent: What Reproduces? [A Reproduction]

Five claims tested with 2,280 pipeline fits, five full CIFAR-10 runs, and fail-closed T4/A10G evidence.

Independent verification

ICML 2026 Agent Reproduction Challenge · OpenReview kcPPWaoegr

VERDICT Fresh GPU evidence supports diversity, reward logic, and full-data CIFAR measurement; missing agent assets block the matched 7B-versus-frontier comparison under the paper's exact evaluation harness, prompts, and training configuration.

2,280

fresh pipeline fits on T4

86.69%

five-seed CIFAR proxy mean

5,000

random diversity controls

1 / 5

headline claim fully blocked

1 Bottom line ★ HEADLINE

This is a **partial reproduction with fresh execution**: Claims 2–5 receive new measurements; Claim 1 remains source-only because the trained agent and harness are absent.

9 → 10

scaled train-to-held-out tasks

0.898

held-out accuracy

0.848

random-action control

Judge rule: paper values are not fresh measurements.

2 Release gate

Asset	Public?	Effect
Trained checkpoint	No	No agent run
Evaluator	No	No Table 1 rerun
Step-wise RL	No	No training replay
Trajectories	No	No data audit
SFT / eval scripts	Stubs	“To be released”

Official commit 15932e7, re-audited 17 July 2026. Missing assets block exact evidence; they do not falsify the paper.

3 Evidence policy

- **Source-verified:** statement located in v1 or v2.
- **Reproduced:** direct independent executable check.
- **Proxy:** disclosed substitute model, task, or scale.
- **Blocked:** required public assets are absent.

3a Primary sources

- **Paper:** arXiv 2505.23723 v1 and v2.
- **Review:** OpenReview kcPPWaoegr.
- **Code:** bingreeky/ML-Agent at commit 15932e7.

Poster hotspots open the matching Trackio evidence page; archived PDFs and Job outputs are bundled.

4 Claim 1 · Model scale

INCONCLUSIVE / BLOCKED: v1 Table 1 reports the 7B Qwen2.5-based agent above GPT-4o, GPT-4o-mini, and 671B DeepSeek-R1 agents. No new matched comparison is possible from the release. Revision warning: v2 replaces the GPT-4o comparators with GPT-5, Gemini, and Qwen3 agents; the challenge wording is v1-specific.

5 Claim 2 · Generalization

PARTIAL / PROXY: a task-conditioned action ranker trained on nine public tasks and evaluated on ten unseen tasks across the paper's image, text, and tabular modalities.

- T4 Job completed all 2,280 pipeline fits.
- 0.897953 mean versus 0.848451 random choice.
- Still below 0.902477 global-best action.

Boundary: no Qwen2.5-7B EFT or step-wise RL.

6 Claim 3 · CIFAR-10

86.69%

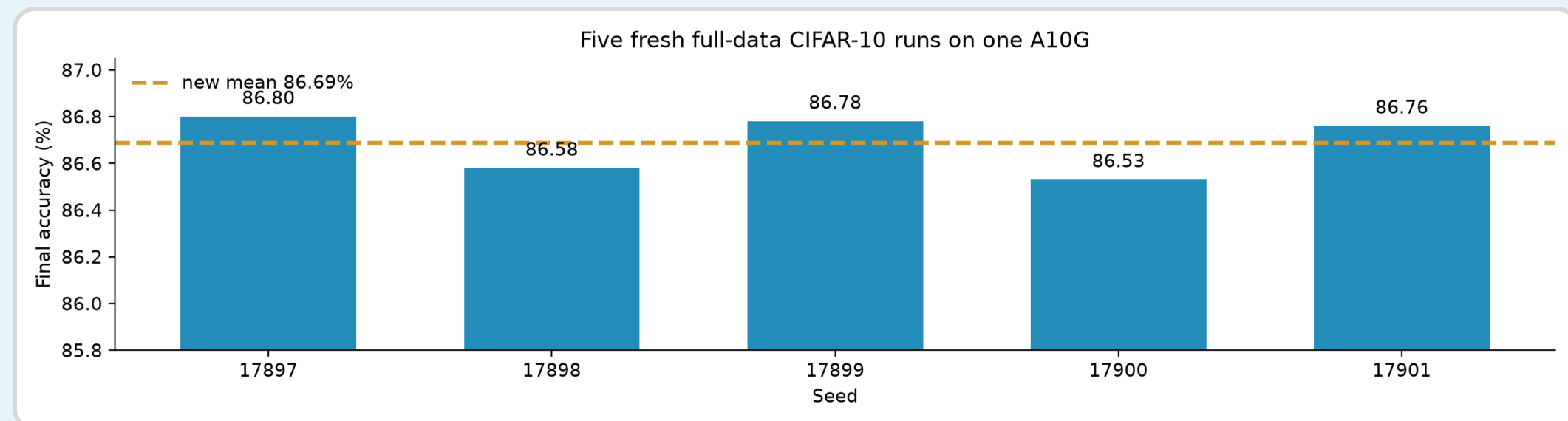
new five-seed mean

86.80%

new best run

±0.11

95% CI half-width, points

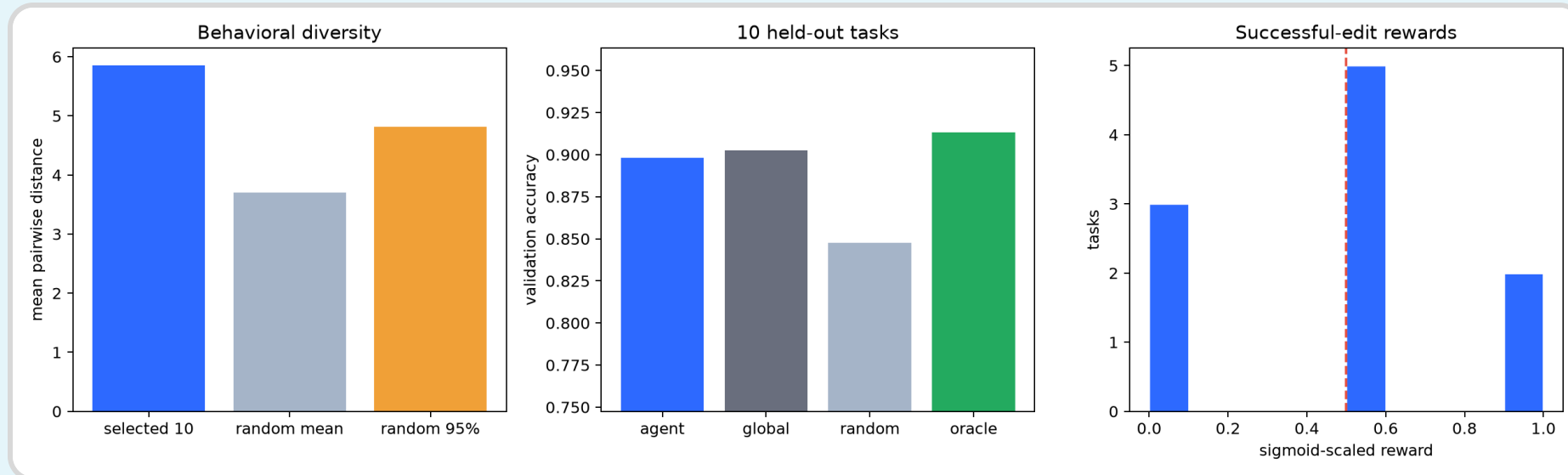


Fresh A10G evidence. Five independent seeds cluster from 86.53% to 86.80%; the dashed line is the new 86.69% mean.

PARTIAL / PROXY: five fresh A10G runs used all 50k train and 10k test images. This CNN is not ML-Agent; comparison to 68.88% / 81.45% is contextual only.

Audit: 3,881 s on one A10G (about USD 1.08); raw results and checksums are Bucket-backed.

7 Claim 4 · Diversity selection ★ PROXY



Independent T4 proxy. 120 executable actions; behavior is the nine-task validation vector, not idea text. Ten maximin actions beat every one of 5,000 random subsets.

5.8477 selected distance versus 3.6949 random mean, 4.8281 random p95, and empirical p = 0.00020.

5.848

selected distance

3.695

random mean

0.00020

empirical p

8 What Claim 4 establishes

- **Fresh actions:** every candidate was fitted on every task.
- **Behavioral distance:** standardized validation vectors.
- **Not established:** author idea pool or downstream RL gain.

A naïve pool-distance rule failed first (p = 0.933); the retained negative run motivated standard maximin farthest-point selection.

Control: selected distance also exceeded the maximum of all 5,000 sampled random subsets.

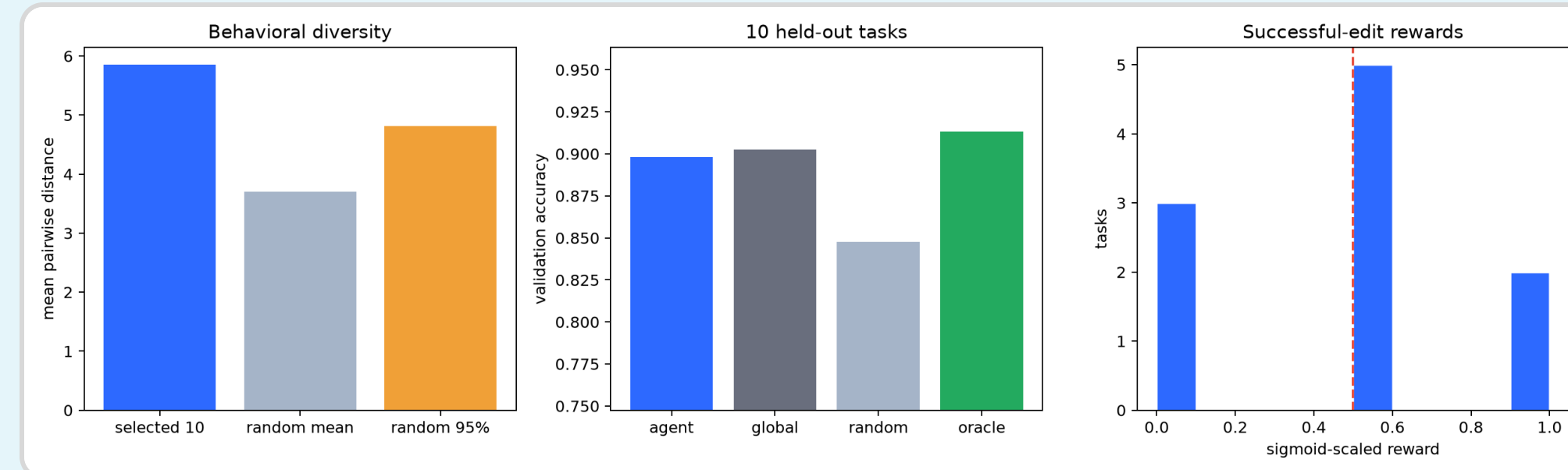
Selection audit: the saved 120 × 9 matrix, fixed seeds, and deterministic ties make the selection independently reproducible.

9 Compute audit

- T4 suite: 243 s running, about USD 0.03.
- A10G CIFAR: 3,881 s running, about USD 1.08.
- Verifier transcripts accompany both successful runs.

Data: a public HF Bucket preserves CIFAR-10 and 20 Newsgroups inputs for reruns.

10 Claim 5 · Reward ★ REPRODUCED



Executable tests. A real invalid estimator, measured p95 latency corner, and ten held-out deltas exercise every v1 branch.

The run preserves **ValueError: not-a-model** feedback and assigns bounded sigmoid rewards to all ten successful edits.

11 Reward rule changed

Trajectory	v1	v2
Invalid / error	0	−1
Corner / non-edit	0.5	0
Improvement	Sigmoid	Linear
Executed edits	10 cases	0 to 1
0	0.5	σ(Δ)
error reward	corner reward	edit reward

Claim 5 reproduces the stated v1 mapping. Author code is absent; v2 changes the rule.

Fail-closed verifier: requires the error string, one corner event, ten bounded sigmoid rewards, and finite deltas.

12 What would unlock a full rerun

- Publish the trained Qwen2.5-7B checkpoint.
- Release evaluator, task environments, and exact prompts.
- Release SFT/RL code plus training trajectories and seeds.

Take-home: Claims 2–5 have fresh evidence at disclosed scope; Claim 1 remains paper-only.

13 Takeaways

CLAIM 1. Blocked by unreleased agent assets.

CLAIMS 2 & 4. T4 proxies beat random controls.

CLAIM 3. Five full CIFAR runs: 86.69% mean.

CLAIM 5. Executed v1 reward branches reproduced.

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